3. REQUIREMENT ANALYSIS

Customer Journey Map

A Customer Journey Map represents the steps a user follows while interacting with the House Rent Management System. It helps in understanding user behavior and improving the system experience.

User Journey Steps

Stage	User Action	System Response	User Goal
Visit Website	User opens the HouseHunt website	Homepage with featured properties is displayed	Explore available rentals
Search Property	User enters location, price range, and property type	System filters and shows relevant listings	Find suitable property
View Details	User clicks on a property	Detailed page with images, price, and description appears	Evaluate property
Register/Login	User creates an account or logs in	Secure authentication is performed	Access personalized features
Contact Owner	User sends inquiry to property owner	Owner receives notification	Get more information
Post Property	Owner lists property with details and images	Listing is saved in database	Rent out property
Admin Moderation	Admin reviews listings	Approves or rejects property posts	Maintain quality and security

This journey ensures that the platform provides a smooth and user-friendly experience for both **tenants and property owners**.

Solution Requirement

The House Rent Management System requires both functional and non-functional requirements.

1. Functional Requirements

Functional requirements describe what the system should do.

User Module

- Users can register and log in securely.
- Users can search properties by location, price range, and property type.
- Users can view property details including images and descriptions.
- Users can contact property owners.

Property Owner Module

- Owners can create and manage property listings.
- Owners can upload images and descriptions of properties.
- Owners can edit or remove their listings.

Admin Module

- Admin can monitor all users and listings.
- Admin can approve or reject property listings.
- Admin can remove inappropriate or fake listings.

Search and Filter System

- Filter properties by:
  - Location
  - Price range
  - Property type (Apartment, House, Villa)

Authentication System

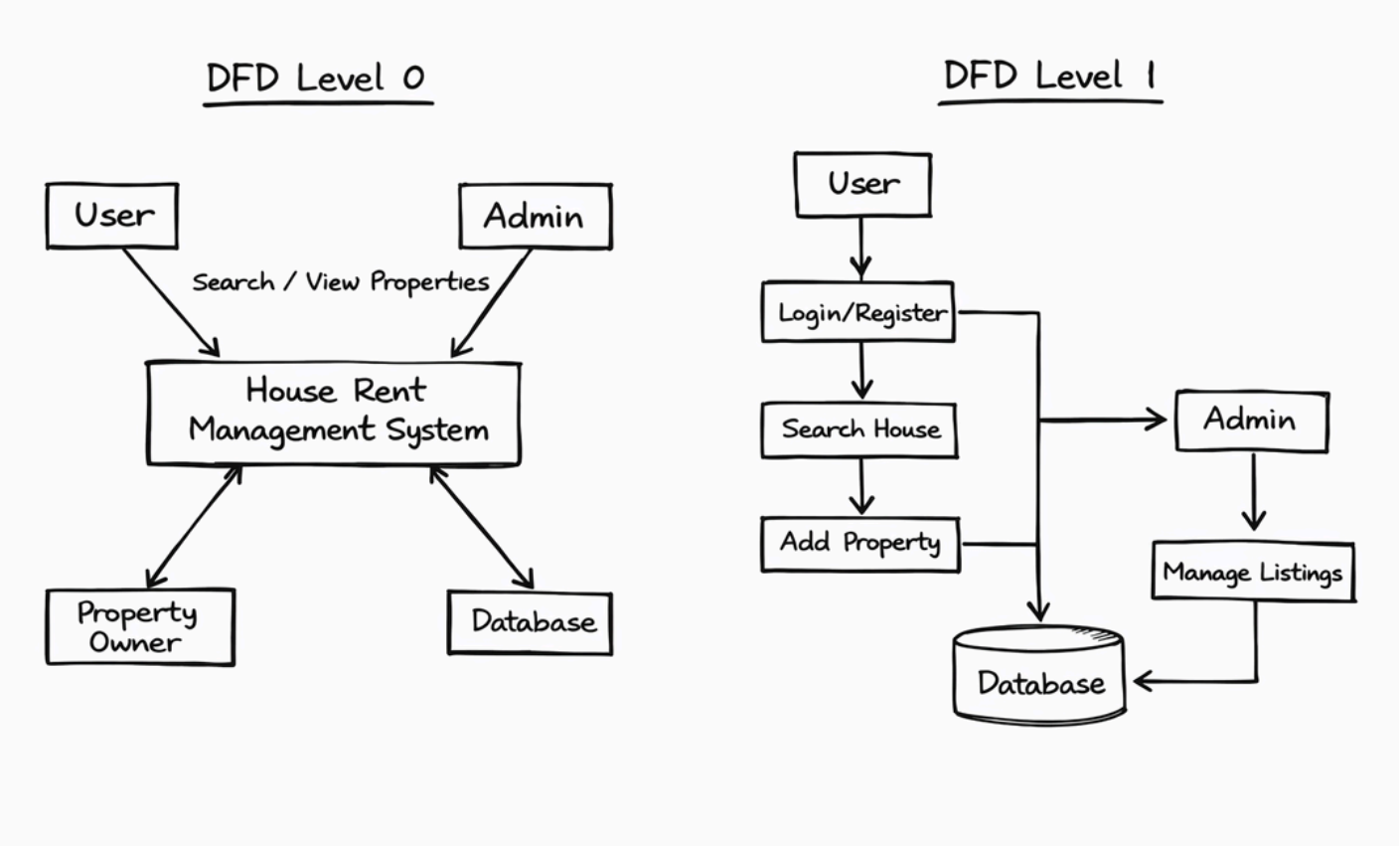
- Secure login using JWT authentication.
- Role-based access like User / Owner / Admin.

2. Non-Functional Requirements

Non-functional requirements describe how the system performs.

- Performance: System should load property listings quickly.
- Security: Secure authentication and data protection.
- Scalability: Should handle increasing number of users and listings.
- Usability: Easy-to-use and responsive interface.
- Reliability: System should be available with minimal downtime.
- Compatibility: Works across devices like mobile, tablet, desktop.

Data Flow Diagram



DFD Level 0

- **User:** Searches and views available rental houses through the system.
- **Property Owner:** Adds property details such as location, rent price, and description.
- **Admin:** Monitors and controls the overall system activities.
- **House Rent Management System:** The central process that connects users, owners, and admin.
- **Database:** Stores all system information including users and property data.

DFD Level 1

- **Login/Register:** Allows users to create an account or log in to the system.
- **Search House:** Users search for available rental houses based on their preferences.
- **Add Property:** Property owners upload new property information to the system.
- **Manage Listings:** Admin verifies, updates, or removes property listings.
- **Database:** Stores and retrieves all data related to users and properties.

Tech Stack

The project is developed using the MERN Stack, which includes the following technologies:

Frontend

- React.js
- HTML5
- CSS3
- JavaScript
- Responsive UI design

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose for database schema modeling

Authentication

- JWT (JSON Web Token)
- bcrypt for password hashing

Development Tools

- Visual Studio Code
- GitHub for version control
- Postman for API testing